Johnson C. Smith University

Precalculus I | MTH 137 | Sections B | Spring 2013

"Every act of conscious learning requires the willingness to suffer an injury to one's self-esteem. That is why young children, before they are aware of their own self-importance, learn so easily; and why older persons, especially if vain or important, cannot learn at all."
–Thomas Szasz, author, professor of psychiatry (1920-)

Instructor Information

Name: Dr. James Ashe
Office location: 307 Davis Hall (back); office hours will also be held in the Math Lab, EDU 115.
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Office hours: MWF 10–11am and 12-1pm; By appointment only, MW 8-9am and 2-3pm.

Course Description

This is a **3 hour** course; the first of a two-course sequence to prepare students for calculus (MTH 231). The course is intended for students who plan to major or minor in mathematics, science, engineering, or computer science. Topics covered: Equations, inequalities, and modeling; functions and graphs; exponents and radicals; polynomial and rational functions; exponential and logarithmic functions; and conic sections. Prerequisite: MTH 131 or by placement or consent of Department.

Instructional methods will primarily consist of lectures and in-class discussions, however other methods may be adopted according to the particular needs of the class.

Course Materials

Textbook/MyMathLab: *Precalculus: Functions and Graphs*, fourth edition, Mark Dugopolski ISBN ISBN-10: 0321837584
A MyMathLab access code is required, and the Ebook/textbook is recommended. New textbooks and Ebooks come packaged with a MyMathLab code.
A MyMathLab code (with Ebook) can be purchased and accessed here: *www.mymathlab.com*

Calculator: A TI-83, TI-83+, or TI-84+ are allowed and supported. Any other calculator must first be approved by the instructor. Students should bring their calculator to class.

Grades

Evaluation: Grades will be taken from homework, quizzes, and Exams as follows:
HW/Quizzes= 20%; Exams 1, 2, 3, and 4 = 15% each; Final Exam = 20%.

Grading scale: $\mathbf{A} \geq 90 > \mathbf{B} \geq 80 > \mathbf{C} \geq 70 > \mathbf{D} \geq 60 > \mathbf{F}$

HW/Quizzes: HW/Quizzes will consist of any graded assignment which is not an exam. This will include traditional quizzes, MyMathLab assignments, take home assignments, and in-class projects. At least the two lowest quiz grades will be dropped.
Exams: Dates for exams will announced at least one week in advance.
Final Exam: The final will be a cumulative exam. The final will be given in our normal class room on Thursday, May 16 from 8am-1pm; see the exam schedule: www.jcsu.edu/happenings/exam-schedule/spring-exam. *If beneficial, the final exam score will replace the lowest exam score.*

Class Policies

JCSU policies regarding attendance, academic integrity, grades, and change of enrollment will be enforced. It is the student's responsibility to be aware of these policies and of related deadlines as well as any announcements or syllabus adjustments (written or oral) made in class. Please see the student handbook for details, jcsu.edu/docs/handbook.pdf.

Attendance: Class attendance is required for all JCSU students. Each student is allowed as many hours of absence per term as credit hours(s) received (not to exceed 4) for the class. Students who exceed the maximum number of absences may receive a failing grade for the course.

Students who may miss classes while representing the University in an official capacity are exempt from regulations governing absences. However, absence from class for official University business does not relieve the student from responsibility for any class assignments that may be missed during the period of absence. In such cases, *it is the student's responsibility to make arrangements to make up any work that will be missed before the assignment/quiz/exam is missed.*

Missed HW/quizzes: There will be no make-ups for in-class quizzes unless the class was missed for a University sponsored event, and advance notice of the absence was given by the student. In this case the assignment will be excused or a ma*Precalculus: Functions and Graphs*, fourth edition, Mark Dugopolski ISBN ISBN-10: 0321837584ke-up will be given depending on circumstances. No late homework will be accepted.
Missed exams: Missed exams may be made up at the discretion of the Instructor. (1) you must notify me by email or phone within 24 hours of the missed exam AND (2) you must have no more than 3 unexcused absences in the class. If BOTH of these conditions are met, a make-up will be given. However unless a documented excuse is provided, 10% will automatically be deducted from the make-up. If you will be gone for any reason, it will be *your* responsibility to inform me at least a week in advance to schedule to take the exam early.

Student support services: If you need course adaptations or accommodations for a documented disability please contact the Office of Disability Services.
jcsu.edu/directory/federal-requirements/general-institutional-information/jcsu-disability-services,
phone: (704) 398-1116.

The Office of Counseling Services provides confidential intakes and assessments, personal counseling and referrals to appropriate resources to all enrolled students on a walk-in or appointment basis.
jcsu.edu/admissions/future_students/meet_your_counselor, phone (704) 378-1044.

Course Content

Chapter 1: Equations, Inequalities, and Modeling
1.1 Equations on One Variable
1.2 Constructing Models to Solve Problems
1.3 Equations and Graphs in Two Variables
1.4 Linear Equations in Two Variables
1.6 Complex Numbers
1.7 Quadratic Equations
1.8 Linear and Absolute Value Inequalities

Chapter 2: Functions and Graphs
2.1 Basics of Functions and Their Graphs
2.2 Graphs of Relations and Functions
2.3 Linear Functions and Slope
2.4 Operations with Functions
2.5 Inverse Functions
2.6 Constructing Functions with Variation

Chapter 3: Polynomial and Rational Functions
3.1 Quadratic Functions and Inequalities
3.2 Zeros of Polynomial Functions
3.3 The Theory of Equations
3.4 Miscellaneous Equations
3.5 Graphs of Polynomial Functions
3.6 Rational Functions and Inequalities

Chapter 4: Exponential and Logarithmic Functions
4.1 Exponential Functions and Their Applications
4.2 Logarithmic Functions and Their Applications
4.3 Rules of Logarithms
4.4 More Equations and Applications

Chapter 10: The Conic Sections
10.1 The Parabola
10.2 The Ellipse and the Circle
10.3 The Hyperbola
10.4 Rotation of Axes
10.5 Polar Equations of the Conics.

While I do not foresee any significant changes to the items in this syllabus during the semester, I reserve the right to make appropriate changes. Students will be notified of any such changes in writing.

JCSU's Fall 2012 academic calendar can be found here: jcsu.edu/happenings/academic-calendar